

4.4 GAUSS–JORDAN ELIMINATION METHOD

The Gauss–Jordan elimination method is a procedure for solving a system of linear equations, $[a][x] = [b]$. In this procedure, a system of equations that is given in a general form is manipulated into an equivalent system of equations in *diagonal* form (see Section 4.1.1) with normalized elements along the diagonal. This means that when the diagonal form of the matrix of the coefficients, $[a]$, is reduced to the identity matrix, the new vector $[b']$ is the solution. The starting point of the procedure is a system of equations given in a general form (the illustration that follows is for a system of four equations):

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Figure 4-17: Matrix form of a system of four equations.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b'_1 \\ b'_2 \\ b'_3 \\ b'_4 \end{bmatrix}$$

Figure 4-18: Matrix form of the equivalent system after applying the Gauss–Jordan method.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 &= b_1 & (4.18a) \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 &= b_2 & (4.18b) \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 &= b_3 & (4.18c) \\ a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 &= b_4 & (4.18d) \end{aligned} \quad (4.18)$$

The matrix form of the system is shown in Fig. 4-17. In the Gauss–Jordan elimination method, the system of equations is manipulated to have the following diagonal form:

$$\begin{aligned} x_1 + 0 + 0 + 0 &= b'_1 & (4.19a) \\ 0 + x_2 + 0 + 0 &= b'_2 & (4.19b) \\ 0 + 0 + x_3 + 0 &= b'_3 & (4.19c) \\ 0 + 0 + 0 + x_4 &= b'_4 & (4.19d) \end{aligned} \quad (4.19)$$

The matrix form of the equivalent system is shown in Fig. 4-18. The terms on the right-hand side of the equations (column $[b']$) are the solution. In matrix form, the matrix of the coefficients is transformed into an identity matrix.

Gauss–Jordan elimination procedure

The Gauss–Jordan elimination procedure for transforming the system of equations from the form in Eqs. (4.18) to the form in Eqs. (4.19) is the same as the Gauss elimination procedure (see Section 4.2), except for the following two differences:

- The pivot equation is normalized by dividing all the terms in the equation by the pivot coefficient. This makes the pivot coefficient equal to 1.
- The pivot equation is used to eliminate the off-diagonal terms in **ALL** the other equations. This means that the elimination process is applied to the equations (rows) that are above and below the pivot equation. (In the Gaussian elimination method, only elements that are below the pivot element are eliminated.)

When the Gauss–Jordan procedure is programmed, it is convenient and more efficient to create a single matrix that includes the matrix of coefficients $[a]$ and the vector $[b]$. This is done by appending the vec-

tor $[b]$ to the matrix $[a]$. The augmented matrix at the starting point of the procedure is shown (for a system of four equations) in Fig. 4-19a. At the end of the procedure, shown in Fig. 4-19b, the elements of $[a]$ are replaced by an identity matrix, and the column $[b']$ is the solution.

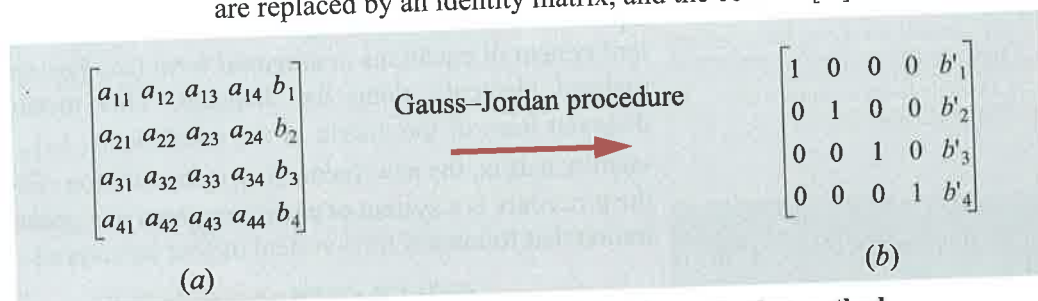


Figure 4-19: Schematic illustration of the Gauss-Jordan method.

The Gauss-Jordan method can also be used for solving several systems of equations $[a][x] = [b]$ that have the same coefficients $[a]$ but different right-hand-side vectors $[b]$. This is done by augmenting the matrix $[a]$ to include all of the vectors $[b]$. In Section 4.6.2 the method is used in this way for calculating the inverse of a matrix.

The Gauss-Jordan elimination method is demonstrated in Example 4-4 where it is used to solve the set of equations solved in Example 4-1.

Example 4-4: Solving a set of four equations using Gauss-Jordan elimination.

Solve the following set of four equations using the Gauss-Jordan elimination method.

$$\begin{aligned} 4x_1 - 2x_2 - 3x_3 + 6x_4 &= 12 \\ -6x_1 + 7x_2 + 6.5x_3 - 6x_4 &= -6.5 \\ x_1 + 7.5x_2 + 6.25x_3 + 5.5x_4 &= 16 \\ -12x_1 + 22x_2 + 15.5x_3 - x_4 &= 17 \end{aligned}$$

SOLUTION

The solution is carried out by using the matrix form of the equations. In matrix form, the system is:

$$\begin{bmatrix} 4 & -2 & -3 & 6 \\ -6 & 7 & 6.5 & -6 \\ 1 & 7.5 & 6.25 & 5.5 \\ -12 & 22 & 15.5 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 12 \\ -6.5 \\ 16 \\ 17 \end{bmatrix}$$

For the numerical procedure, a new matrix is created by augmenting the coefficient matrix to include the right-hand side of the equation:

$$\begin{bmatrix} 4 & -2 & -3 & 6 & 12 \\ -6 & 7 & 6.5 & -6 & -6.5 \\ 1 & 7.5 & 6.25 & 5.5 & 16 \\ -12 & 22 & 15.5 & -1 & 17 \end{bmatrix}$$

The first pivoting row is the first row, and the first element in this row is the pivot element. The row is normalized by dividing it by the pivot element:

$$\begin{bmatrix} \frac{4}{4} & \frac{-2}{4} & \frac{-3}{4} & \frac{6}{4} & \frac{12}{4} \\ -6 & 7 & 6.5 & -6 & -6.5 \\ 1 & 7.5 & 6.25 & 5.5 & 16 \\ -12 & 22 & 15.5 & -1 & 17 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 & -0.75 & 1.5 & 3 \\ -6 & 7 & 6.5 & -6 & -6.5 \\ 1 & 7.5 & 6.25 & 5.5 & 16 \\ -12 & 22 & 15.5 & -1 & 17 \end{bmatrix}$$

Next, all the first elements in rows 2, 3, and 4 are eliminated:

$$\begin{bmatrix} 1 & -0.5 & -0.75 & 1.5 & 3 \\ -6 & 7 & 6.5 & -6 & -6.5 \\ 1 & 7.5 & 6.25 & 5.5 & 16 \\ -12 & 22 & 15.5 & -1 & 17 \end{bmatrix} \begin{array}{l} \leftarrow -(-6)[1 \ -0.5 \ -0.75 \ 1.5 \ 3] \\ \leftarrow -(1)[1 \ -0.5 \ -0.75 \ 1.5 \ 3] \\ \leftarrow -(-12)[1 \ -0.5 \ -0.75 \ 1.5 \ 3] \end{array} = \begin{bmatrix} 1 & -0.5 & -0.75 & 1.5 & 3 \\ 0 & 4 & 2 & 3 & 11.5 \\ 0 & 8 & 7 & 4 & 13 \\ 0 & 16 & 6.5 & 17 & 53 \end{bmatrix}$$

The next pivot row is the second row, with the second element as the pivot element. The row is normalized by dividing it by the pivot element:

$$\begin{bmatrix} 1 & -0.5 & -0.75 & 1.5 & 3 \\ 0 & \frac{4}{4} & \frac{2}{4} & \frac{3}{4} & \frac{11.5}{4} \\ 0 & 8 & 7 & 4 & 13 \\ 0 & 16 & 6.5 & 17 & 53 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 & -0.75 & 1.5 & 3 \\ 0 & 1 & 0.5 & 0.75 & 2.875 \\ 0 & 8 & 7 & 4 & 13 \\ 0 & 16 & 6.5 & 17 & 53 \end{bmatrix}$$

Next, all the second elements in rows 1, 3, and 4 are eliminated:

$$\begin{bmatrix} 1 & -0.5 & -0.75 & 1.5 & 3 \\ 0 & 1 & 0.5 & 0.75 & 2.875 \\ 0 & 8 & 7 & 4 & 13 \\ 0 & 16 & 6.5 & 17 & 53 \end{bmatrix} \begin{array}{l} \leftarrow -(-0.5)[0 \ 1 \ 0.5 \ 0.75 \ 2.875] \\ \leftarrow -(8)[0 \ 1 \ 0.5 \ 0.75 \ 2.875] \\ \leftarrow -(16)[0 \ 1 \ 0.5 \ 0.75 \ 2.875] \end{array} = \begin{bmatrix} 1 & 0 & -0.5 & 1.875 & 4.4375 \\ 0 & 1 & 0.5 & 0.75 & 2.875 \\ 0 & 0 & 3 & -2 & -10 \\ 0 & 0 & -1.5 & 5 & 7 \end{bmatrix}$$

The next pivot row is the third row, with the third element as the pivot element. The row is normalized by dividing it by the pivot element:

$$\begin{bmatrix} 1 & 0 & -0.5 & 1.875 & 4.4375 \\ 0 & 1 & 0.5 & 0.75 & 2.875 \\ 0 & 0 & \frac{3}{3} & \frac{-2}{3} & \frac{-10}{3} \\ 0 & 0 & -1.5 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -0.5 & 1.875 & 4.4375 \\ 0 & 1 & 0.5 & 0.75 & 2.875 \\ 0 & 0 & 1 & -0.667 & -3.333 \\ 0 & 0 & -1.5 & 5 & 7 \end{bmatrix}$$

Next, all the third elements in rows 1, 2, and 4 are eliminated:

$$\begin{bmatrix} 1 & 0 & -0.5 & 1.875 & 4.4375 \\ 0 & 1 & 0.5 & 0.75 & 2.875 \\ 0 & 0 & 1 & -0.667 & -3.333 \\ 0 & 0 & -1.5 & 5 & 7 \end{bmatrix} \begin{array}{l} \leftarrow -(-0.5)[0 \ 0 \ 1 \ -0.667 \ -3.333] \\ \leftarrow -(0.5)[0 \ 0 \ 1 \ -0.667 \ -3.333] \\ \leftarrow -(-1.5)[0 \ 0 \ 1 \ -0.667 \ -3.333] \end{array} = \begin{bmatrix} 1 & 0 & 0 & 1.5417 & 2.7708 \\ 0 & 1 & 0 & 1.0833 & 4.5417 \\ 0 & 0 & 1 & -0.667 & -3.333 \\ 0 & 0 & 0 & 4 & 2 \end{bmatrix}$$

The next pivot row is the fourth row, with the fourth element as the pivot element. The row is normalized by dividing it by the pivot element:

$$\begin{bmatrix} 1 & 0 & 0 & 1.5417 & 2.7708 \\ 0 & 1 & 0 & 1.0833 & 4.5417 \\ 0 & 0 & 1 & -0.667 & -3.333 \\ 0 & 0 & 0 & \frac{4}{4} & \frac{2}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1.5417 & 2.7708 \\ 0 & 1 & 0 & 1.0833 & 4.5417 \\ 0 & 0 & 1 & -0.667 & -3.333 \\ 0 & 0 & 0 & 1 & 0.5 \end{bmatrix}$$

Next, all the fourth elements in rows 1, 2, and 3 are eliminated

$$\begin{bmatrix} 1 & 0 & 0 & 1.5417 & 2.7708 \\ 0 & 1 & 0 & 1.0833 & 4.5417 \\ 0 & 0 & 1 & -0.667 & -3.333 \\ 0 & 0 & 0 & 1 & 0.5 \end{bmatrix} \begin{array}{l} \leftarrow -(1.5417)[0 \ 0 \ 0 \ 1 \ 0.5] \\ \leftarrow -(1.0833)[0 \ 0 \ 0 \ 1 \ 0.5] \\ \leftarrow -(-0.667)[0 \ 0 \ 0 \ 1 \ 0.5] \end{array} = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & 0.5 \end{bmatrix}$$

The solution is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -3 \\ 0.5 \end{bmatrix}$$

The Gauss–Jordan elimination method with pivoting

It is possible that the equations are written in such an order that during the elimination procedure a pivot equation has a pivot element that is equal to zero. Obviously, in this case it is impossible to normalize the pivot row (divide by the pivot element). As with the Gauss elimination method, the problem can be corrected by using pivoting. This is left as an exercise in Problem 4.24.

4.5 LU DECOMPOSITION METHOD

Background

The Gauss elimination method consists of two parts. The first part is the elimination procedure in which a system of linear equations that is given in a general form, $[a][x] = [b]$, is transformed into an equivalent system of equations $[a'][x] = [b']$ in which the matrix of coefficients $[a']$ is upper triangular. In the second part, the equivalent system is solved by using back substitution. The elimination procedure requires many mathematical operations and significantly more computing time than the back substitution calculations. During the elimination procedure, the matrix of coefficients $[a]$ and the vector $[b]$ are both changed. This means that if there is a need to solve systems of equations that have the same left-hand-side terms (same coefficient matrix $[a]$) but different right-hand-side constants (different vectors $[b]$), the elimination procedure has to be carried out for each $[b]$ again. Ideally, it would be better if the operations on the matrix of coefficients $[a]$ were